Slope comp scratchpad

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$$\alpha = \frac{m_{cmp} - m_d}{m_{cmp} + m_c}$$

$$\beta = \frac{m_c + m_d}{m_c + m_{cmp}}$$

$$G_{pk} = \frac{1 - \alpha}{1 + \alpha}$$

$$G_{pk} = \frac{1}{2D\left(\dfrac{m_{cmp}}{m_d} - 1\right) + 1}$$

$$m_{cmp}(D) = \frac{1}{D}\left(\frac{1 - G_{pk}}{2G_{pk}}\right)m_d + m_d$$

$$I_{cmp\_NL}(D) = T_{SW}\int_0^D m_{cmp}(D)dD$$

$$I_{cmp\_NL}(D) = m_d T_{SW}\left(\frac{1 - G_{pk}}{2G_{pk}}\ln(D) + D + \frac{1 - G_{pk}}{2G_{pk}} + 2\ln\left(\frac{G_{pk} - 1}{2G_{pk}}\right)\right)$$

$$I_{cmp\_LIN}(D) = m_{cmp} D T_{SW}$$

$$I_{cmp\_LIN0}(D) = \begin{cases} 0 & , m_{cmp} D T_{SW} \leq I_{OFST} \\ m_{cmp} D T_{SW} - I_{OFST} & , m_{cmp} D T_{SW} > I_{OFST} \end{cases}$$